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NEAR AND FAR-FIELD DEFORMATION FROM THE 2023 TURKEY EARTHQUAKES USING GNSS DATA.

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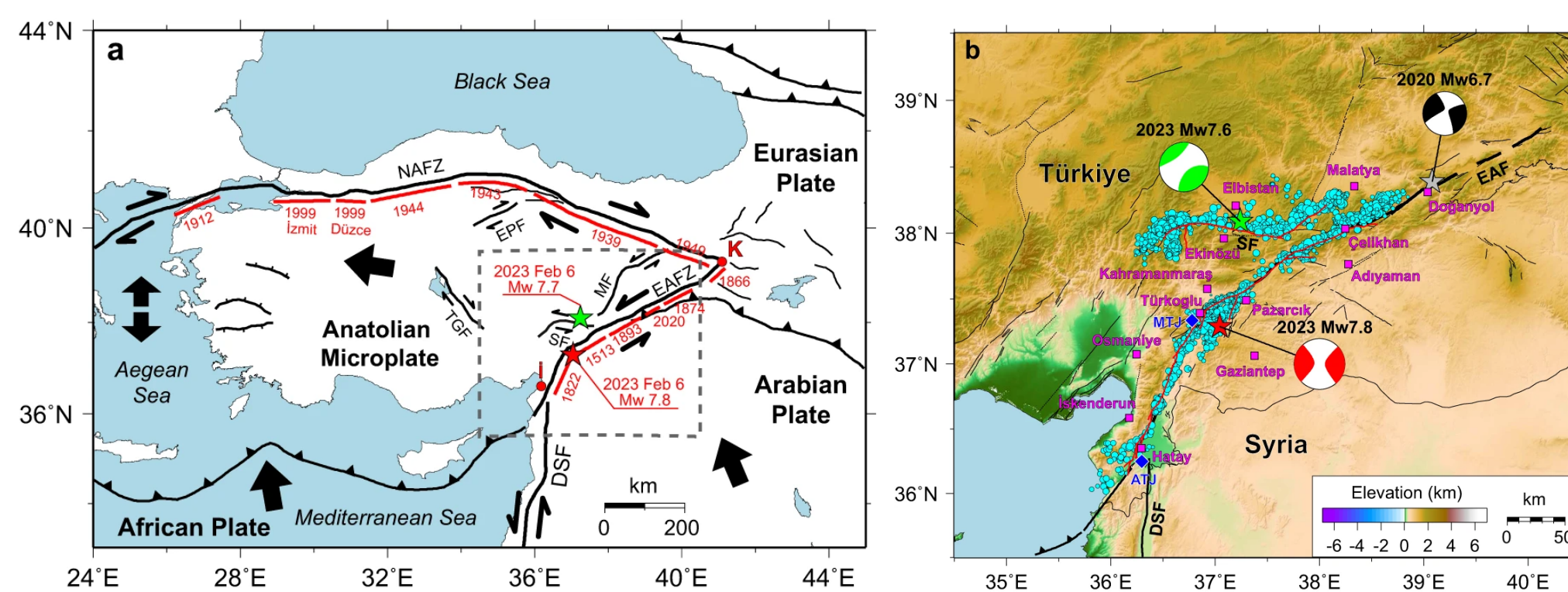
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INTRODUCTION

On February 6th, 2023 at 01:17:35 UTC the M_w 7.8 Nurdağı-Pazarçık earthquake nucleated ~ 15 km southeast of the mapped trace of the East Anatolian Fault Zone. Relocations place the hypocenter at (37.0234° E, 37.2444° N, depth=12 km) and analyses of teleseismic data show a left-lateral source mechanism on a vertical or near vertical fault. A vigorous aftershock sequence followed and a little over 9 hours after the first event, at 10:24:49 UTC, the M_w 7.6 [1]

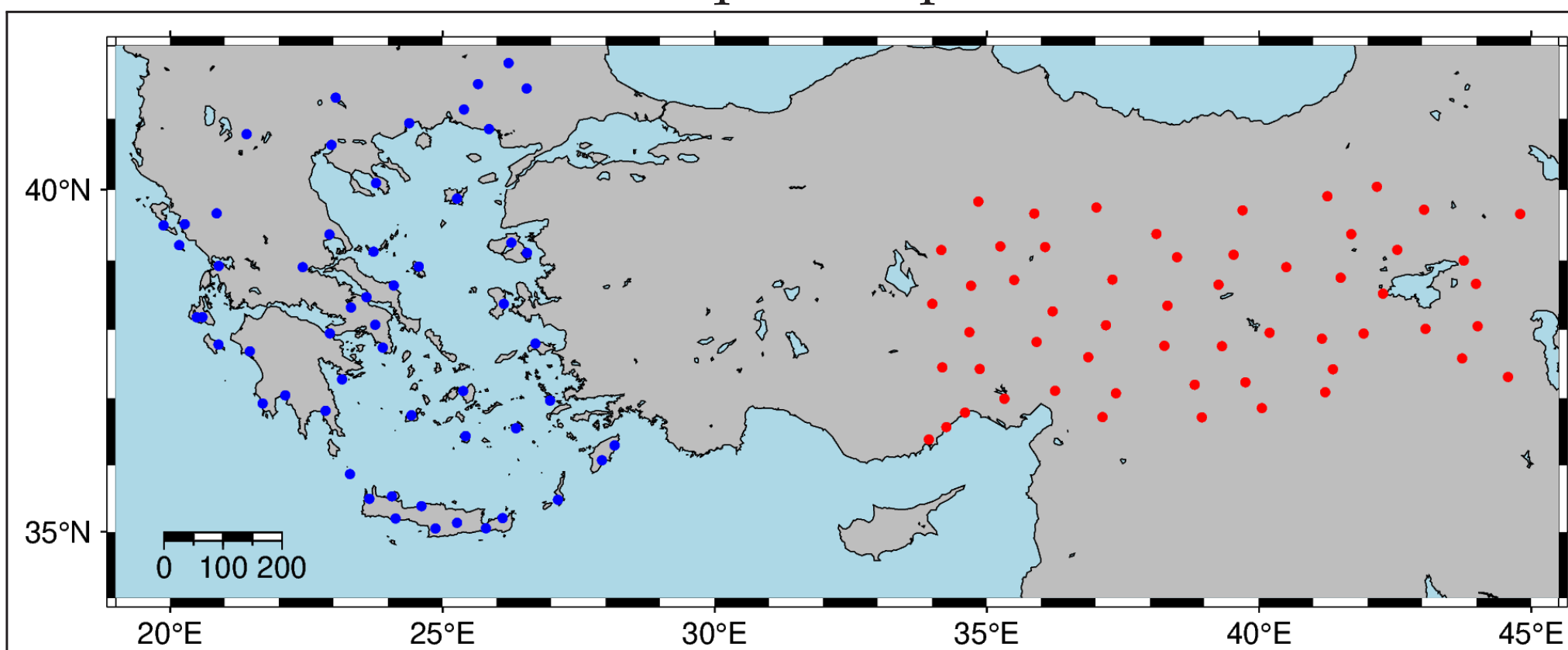


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GNSS STATIONS

41 continuously operating stations located near the earthquake sequence in southwest Turkey (<500km distance from epicentre), and an additional 54 far-field stations distributed across the Aegean Sea and mainland Greece in a distance of 500 to 1000km from the two earthquakes epicentre.



Acknowledgments to Metrica SA for the free availability of the 1Hz GNSS data of HxGN SmartNet for this study, and TUSAGA-Aktif provide GNSS data of Turkish GNSS stations



REFERENCES

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- [2] Chengli Liu, Thorne Lay, Rongjiang Wang, Tuncay Taymaz, Zujun Xie, Xiong Xiong, Tahir Serkan Irmak, Metin Kahraman, and Ceyhan Erman. Complex multi-fault rupture and triggering during the 2023 earthquake doublet in southeastern türkiye. *Nature Communications*, 14(1), September 2023.
- [3] Jianghui Geng, Xingyu Chen, Yuanxin Pan, Shuyin Mao, Chenghong Li, Jinning Zhou, and Kunlun Zhang. Pride ppp-ar: an open-source software for gps ppp ambiguity resolution. *GPS Solutions*, 23(4), July 2019.



DATA ANALYSIS

Data from all available operating stations were analysed.

Processing scheme:

- Precise Point Positioning with Ambiguity Resolution using PRIDE-PPPAR Software [3].
- Satellite systems: GPS / GLO / GAL / BDS-2/3
- Elevation angle: 7 deg
- Kinematic mode
- Reference Frame: IGS20
- Final IGS Products for satellite
- ANTEX File: IGS20_2247
- Tropospheric modeling: VMF1

TS1

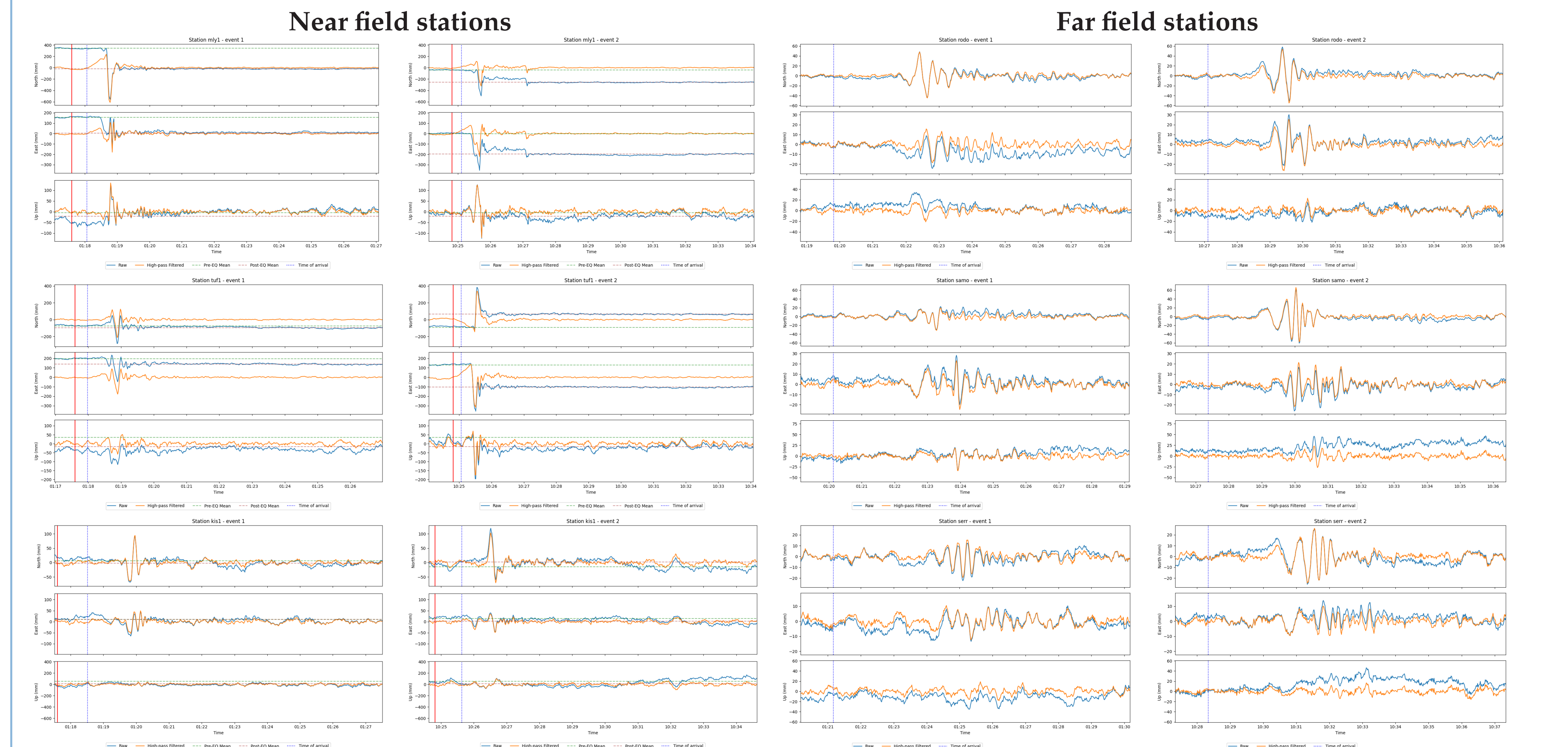
1. Pre-processing and Filtering of GNSS Data

Low-frequency noise in the GNSS data—likely caused by multipath effects, unmitigated tropospheric delays, and other sources—was suppressed using a high-pass Butterworth filter with a cut-off frequency of 0.1 Hz. This filter effectively removed slow variations without significantly affecting any potential dynamic displacements. Since the GNSS stations were located more than 800 km from the earthquake epicenters, no co-seismic displacements were expected.

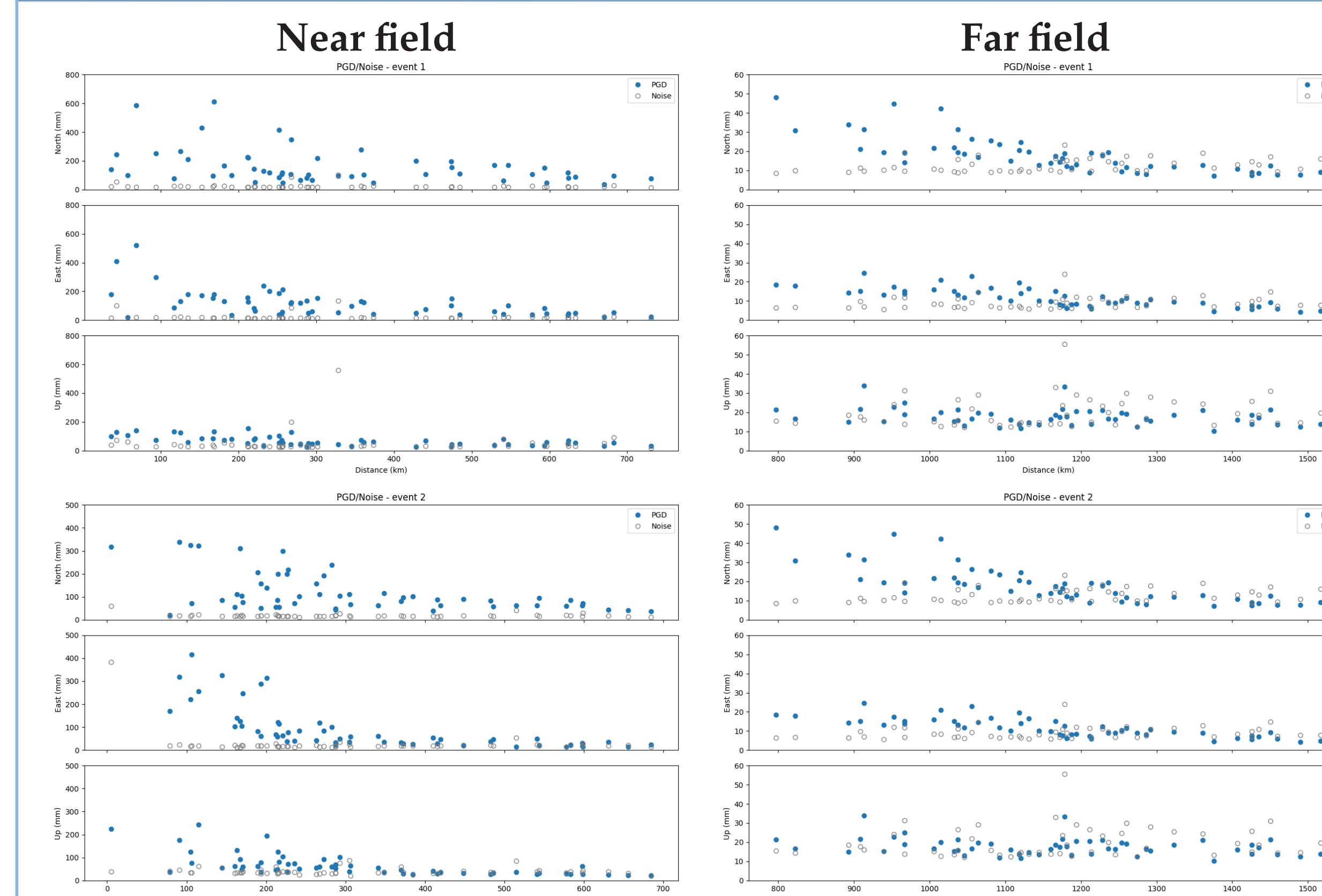
2. Transient Response Detection and Noise Estimation

The transient seismic response window for each GNSS station was defined as starting at the arrival time of the P-waves—calculated assuming a velocity of 6 km/s—and spanning 400 seconds to ensure the full signal was captured. Peak Ground Displacement (PGD) in the neu components was identified as the maximum value within this interval. To estimate the noise level, we calculated the standard deviation of the filtered neu time series over a 10-minute window prior to the first seismic event. A conservative threshold of 3σ was used to distinguish real transient responses from background noise.

TIME SERIES ANALYSIS



DISCUSSION



From the first analysis of the PGD values the following points are observed:

- For both seismic events, in the northing component is observed the highest values of PGD indicating that the this direction is affected by the propagation of the seismic waves.
- The second seismic event has larger far-field impact probably due to the geometry of the seismic fault and the directivity of the seismic waves propagation.
- The Up component is not affected by the seismic events potentially due to the small impact that the S-waves may have for far-field stations.

FUTURE RESEARCH

- Investigate the far field of the directivity effect of the propagation seismic waves through the PGD of the GNSS stations.
- Investigate the frequency context of the seismic waves and its potential variation with the distance from the epicentre.
- Estimate of the seismic waves velocity as detected through the network GNSS stations displacement.

CONTACT INFORMATION

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